

Fundamental scaling laws of water-window X-rays from free-electron-driven van der Waals structures

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Water-window X-rays are crucial in medical and biological applications, enabling the natural-contrast imaging of biological cells without external staining. However, water-window X-ray sources with bespoke photon energies—needed in high-contrast imaging—remain challenging to obtain, except at large synchrotron facilities. Here we address this challenge by demonstrating tabletop, water-window X-ray generation from free-electron-driven van der Waals materials, enabling the continuous tuning of photon energies across the entire water-window regime. Additionally, we present a truly predictive theoretical framework combining first-principles electromagnetism with Monte Carlo simulations to accurately predict the photon flux and brightness in absolute quantities. We obtain fundamental scaling laws for the tunable photon flux, matching the experimental results and providing a way to design powerful emitters based on free-electron-driven quantum materials. We show that we can potentially achieve photon fluxes needed for imaging and spectroscopy applications (over 10^8 photons s^{-1} on the sample—verified by our framework based on our experimentally achieved fluxes of about 10^3 photons s^{-1} using ~50 nA current). Importantly, our theory highlights the critical role played by the large mean free paths and interlayer atomic spacings unique to van der Waals structures, showing the latter's advantages over other materials in generating water-window X-rays.

The water-window regime is a part of the X-ray spectrum in which water is transparent, but carbon is absorptive. In this regime, we may thus observe biological cells and its organelles in a non-destructive manner¹, as cells are composed of more than 70% water in which various carbon-based structures of interest are suspended. Continuously tunable water-window X-ray sources—referring to water-window X-ray sources whose photon energies can be arbitrarily specified—are highly sought after as they can be used to enhance the contrast of cellular images without the aid of staining agents. This enhanced contrast is obtained by leveraging the

fact that different materials have different absorption spectra: as a result, images taken of the same sample at different photon energies will have differences that can be exploited to enhance the image resolution. For example, imaging with photon energies below and above the absorption edges can produce an elemental map with improved contrast for specific elements such as calcium (Ca) and oxygen^{2,3}—such detection of elemental composition is vital for diagnosing diseases in patients.

Modern water-window microscopes rely on synchrotrons that produce continuously tunable X-rays, allowing the photon energy to

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be arbitrarily specified. However, the large size of synchrotrons limits their accessibility, fuelling the development of efficient tabletop water-window microscopes^{1,4–7}. The first of these laboratory-scale microscopes was the soft X-ray transmission microscope at the Royal Institute of Technology in Stockholm⁸, which was based on a liquid-jet high-brightness laser plasma source operating at a fixed wavelength of 2.58 nm. Other tabletop water-window microscopes utilize electron-impact X-ray tubes^{5–7,9} that emit characteristic X-rays, with fixed wavelengths, within the water-window regime. Water-window X-ray sources based on high-harmonic generation^{10–14} enable the generation of coherent and ultrafast water-window X-ray pulses on a tabletop scale, although with the requirement of high-intensity lasers.

Here we introduce a versatile water-window X-ray source based on free-electron-driven van der Waals (vdW) materials. We show that our source is highly complementary to existing water-window X-ray sources, with distinct advantages especially when continuous energy tunability and compactness are important. It should also be noted that the water-window regime, where our theories and experiments take place, constitutes a previously unexplored regime in light emission via free-electron-driven vdW materials. To our knowledge, our findings represent the lowest peak photon energies to be experimentally generated via our X-ray generation mechanisms of choice, namely, parametric X-ray radiation (PXR) and coherent bremsstrahlung (CB). Additionally, we theoretically obtain and experimentally reveal the fundamental scaling laws of the emitted photon flux. In the process, we present a truly predictive theoretical framework that combines first-principles electromagnetism and Monte Carlo simulations to accurately predict the output photon flux and brightness of free-electron-driven quantum materials in absolute numbers. Our framework shows that the tunable photon flux increases linearly with current and sub-linearly with crystal thickness. With an experimental flux of $\sim 10^3$ photons s^{-1} achieved at ~ 50 nA current, we show that we can potentially achieve fluxes of over 10^8 photons s^{-1} on a sample through scaling with the incident current, sufficient for a broad range of laboratory-based imaging applications. We demonstrate the ability to arbitrarily specify the photon energy peak by varying the tilt angle of the crystal and/or the driving electron energy, resulting in an output photon energy peak that can be continuously tuned across the water-window regime. Our framework also reveals the distinct advantages of using vdW materials compared with conventional crystalline materials in our scheme: the larger interatomic two-dimensional layer spacing possessed by vdW materials leads to larger mean free paths and electron interaction lengths, resulting in larger X-ray flux emitted in a narrower bandwidth. Our results should pave the way to compact, versatile water-window X-ray sources for more discriminating medical and biological imaging.

Results and analysis

vdW materials have attracted tremendous attention due to their unique properties such as linear energy–momentum dispersion¹⁵, massive intrinsic charge mobility^{16,17}, extreme electromagnetic confinement^{18,19}, quantum Hall effects^{20–22}, Van Hove singularities²³ and tunable photon polaritons²⁴. Among their many intriguing possibilities, vdW materials are excellent platforms for nanomaterial-based free-electron-driven X-ray sources^{25–31} due to their high in-plane thermal conductivity³² and strong plasmonic confinement²⁵. Because of the large range of possible chemical combinations that allows control over the precise lattice constants defining the radiation spectrum, vdW materials and heterostructures^{33–40} are tempting platforms for the creation of tunable X-rays^{27,29,30,41} and for studying quantum effects in the free-electron-driven X-ray generation process^{42,43}.

Our study focuses on the generation of continuously tunable water-window X-rays using free-electron interactions with vdW materials, with the process depicted in Fig. 1a. These materials consist of layered two-dimensional atomic planes, with larger interlayer

spacing compared with conventional crystals, enabling us to produce high-quality monochromatic tunable water-window X-rays using a tabletop setup. The X-rays are produced through the combination of two mechanisms, namely, PXR and CB^{44–51}. The theory of tunable X-ray generation via the PXR and CB processes is available in Supplementary Section 1. These X-rays are tunable using three different mechanisms^{29,30}: (1) by varying the incident electron energy, (2) by varying the tilt angle of the crystal and (3) by varying the composition of the vdW material. The two-dimensional atomic layers of the vdW materials act like a nano-grating that diffracts the evanescent field of the incident electrons into propagating photons termed as PXR, whereas the other mechanism, CB, accounts for the X-ray emission by an undulated electron due to a series of periodic interactions with the crystal lattice. The resultant spectrum is the interference between PXR and CB. A combination of these two mechanisms is referred to as parametric coherent bremsstrahlung (PCB) radiation. In our study, these two forms of X-ray radiation (PXR and CB) have identical output X-ray photon energies^{29,46} given by the following equation:

$$E = \hbar \frac{\mathbf{v} \cdot \mathbf{g}}{1 - (\mathbf{v} \cdot \hat{\mathbf{n}}/c)}, \quad (1)$$

where $\hbar$ is the reduced Planck constant, $\mathbf{v}$ is the velocity of an electron, $\mathbf{g}$ is the reciprocal lattice vector, $\hat{\mathbf{n}}$ is the unit vector along the observation direction and c is the speed of light in free space. We define the crystal tilt angle θ_{til} as the angle between the [001] zone axis and the z axis (Fig. 1a). The azimuthal rotation angle of the crystal with respect to the z axis is denoted as ϕ_{til} .

Combining this tunable water-window X-ray source with a range of X-ray optics can enable the X-ray imaging of biological cells in a tabletop setup. A schematic of our proposed X-ray imaging setup is shown in Fig. 1b. The X-rays emitted from the target material are selected by a monochromator, which then directs a monochromatic X-ray beam towards the condenser optics. The condenser optics (for example, an X-ray capillary tube or a zone plate) converges the incoming X-rays and focuses it on the desired sample surface after passing through an aperture. The X-rays transmitted through the sample are then detected by an X-ray charge-coupled device (CCD) camera. An objective lens optics can also be used to focus the image onto the CCD camera. This setup can be easily altered to perform near-edge X-ray absorption fine structure (NEXAFS) spectroscopy by replacing the CCD camera with a spectrometer.

Figure 1b (inset) illustrates a potential application of the water-window X-rays generated by this system. Using X-ray images captured above and below the Ca absorption edge, we can acquire high-contrast images of dense granules (DG) in blood platelet cells through a two-wavelength dichromography technique. The outcome of logarithmic subtraction⁵² between the two X-ray images of the examined phantom blood platelet cell effectively highlights a significant contrast difference between DG and other non-DG, facilitating the accurate detection of DG. This capability holds immense significance in the medical industry and proves invaluable for patient diagnosis³. This technique has shown promising results in accurately detecting DG from tunable water-window X-rays generated in synchrotron facilities. Despite our source not possessing the ultrahigh fluxes of synchrotron-based sources, we show that we are able to obtain high-contrast images for DG detection (2,000-fold enhancement), demonstrating the possibility to perform such diagnoses in a tabletop setup. For further details on the blood platelet cell phantom model studied using our X-ray source, please refer to Supplementary Section 3.

We experimentally and theoretically demonstrate that we can adjust the kinetic energy of incident electrons and tilt angle of the vdW crystal to finely tune the emitted X-ray photon energy within the water-window regime. As shown in Fig. 1a, an electron beam impinges on a vdW material, resulting in the emission of water-window X-rays via

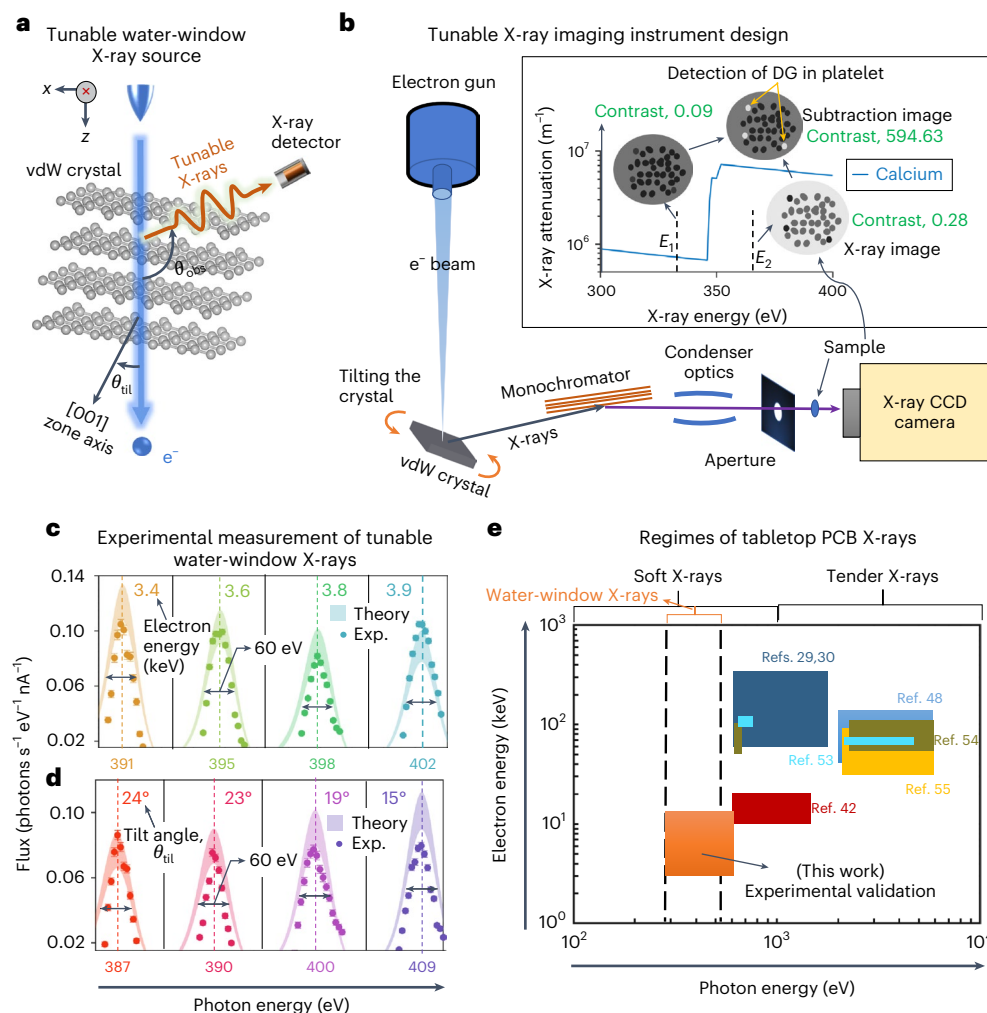


Fig. 1 | Tunable water-window X-rays from free electrons interacting with vdW materials and X-ray imaging instrument design. **a**, Schematic of an incident electron beam passing through a vdW material, generating water-window X-rays via PXR and CB. The observation angle θ_{obs} is 119° . **b**, Schematic of the tabletop water-window X-ray imaging setup. The inset illustrates a potential biological application demonstrating the accurate detection of DG in blood platelet cells using tunable water-window X-rays with properties from our source. The contrast for X-ray images is defined as the flux difference between DG and other non-DG, whereas for the subtraction image, it denotes the difference between the logarithmic subtraction values of the flux from the two X-ray images. **c**, Photon energy spectrum tunability by varying the

incident electron energies in steps of 3.4 keV, 3.6 keV, 3.8 keV and 3.9 keV. The x-axis ticks are 50 eV apart, with each graph partitioned by vertical black lines. **d**, Photon energy spectrum tunability by varying the tilt angle θ_{til} of the crystal by 15° , 19° , 23° and 24° . In **c** and **d**, the experimental data are represented by filled circles and theoretical predictions, by the shaded region. The vertical dashed lines indicate the peak photon energy predicted by equation (1). The error bars represent the standard deviation (s.d.) in the experimental results due to the Poisson statistical error in the photon-counting process of the EDS detector. **e**, X-ray regimes investigated using tabletop electron sources, with particular emphasis on the orange-coloured region to highlight the novelty of this study in an unexplored regime.

PCB radiation. The emitted X-rays are detected by an energy-dispersive X-ray spectroscopy (EDS) detector. The results (Fig. 1c,d) demonstrate a shift in the peak intensity of the emission spectrum as the incident electron energy is varied from 3.4 keV to 3.9 keV (Fig. 1c) and as the tilt angle of the crystal θ_{til} is adjusted in increments of 15° , 19° , 23° and 24° at a constant electron energy of 3.8 keV (Fig. 1d), with peak shift ranging from approximately 385 eV to 410 eV. The tilt angle of the target θ_{til} is accurately determined by fitting the curves' peak intensity using equation (1). The thickness of the graphite crystal used in Fig. 1c,d is 10 nm, which corresponds to 30 atomic layers. The mean free path of 4 keV electrons in graphite is about 5 nm. Details for the experiments presented in Fig. 1 are provided in Supplementary Section 4. In Fig. 1c, the flux ratio of our tunable water-window X-ray source to the incoherent bremsstrahlung flux is about 4. This ratio can be made still larger by using larger electron energies, for instance.

Figure 1e shows the regimes of tabletop X-rays investigated using tabletop electron sources^{29,30,42,48,53–55}, specifically focusing on electron

energies below 1 MeV. Despite the abundance of literature on the relativistic regime^{56–60}, we restrict our attention, in this work, to tabletop electron sources. Until now, expanding this domain to access water-window X-rays had not been feasible. However, in this study, we have successfully demonstrated experimental evidence of accessing the water-window X-rays. This was possible by exploiting the large inter-layer d spacing of vdW materials, which allowed a lower wavelength of photons for relatively higher kinetic energies of electrons compared with conventional crystals.

In numerous X-ray imaging applications, the X-ray photon flux is a critical factor. It is, therefore, important to appreciate the underlying laws governing the scaling of photon flux with various parameters. We have developed a theoretical framework combining first-principles electromagnetism with Monte Carlo simulations to accurately predict the tunable photon flux. Our theoretical framework showing the front-to-end simulation is described in Fig. 2b. The input parameters needed for the flux simulations are listed in Fig. 2a, which consists of

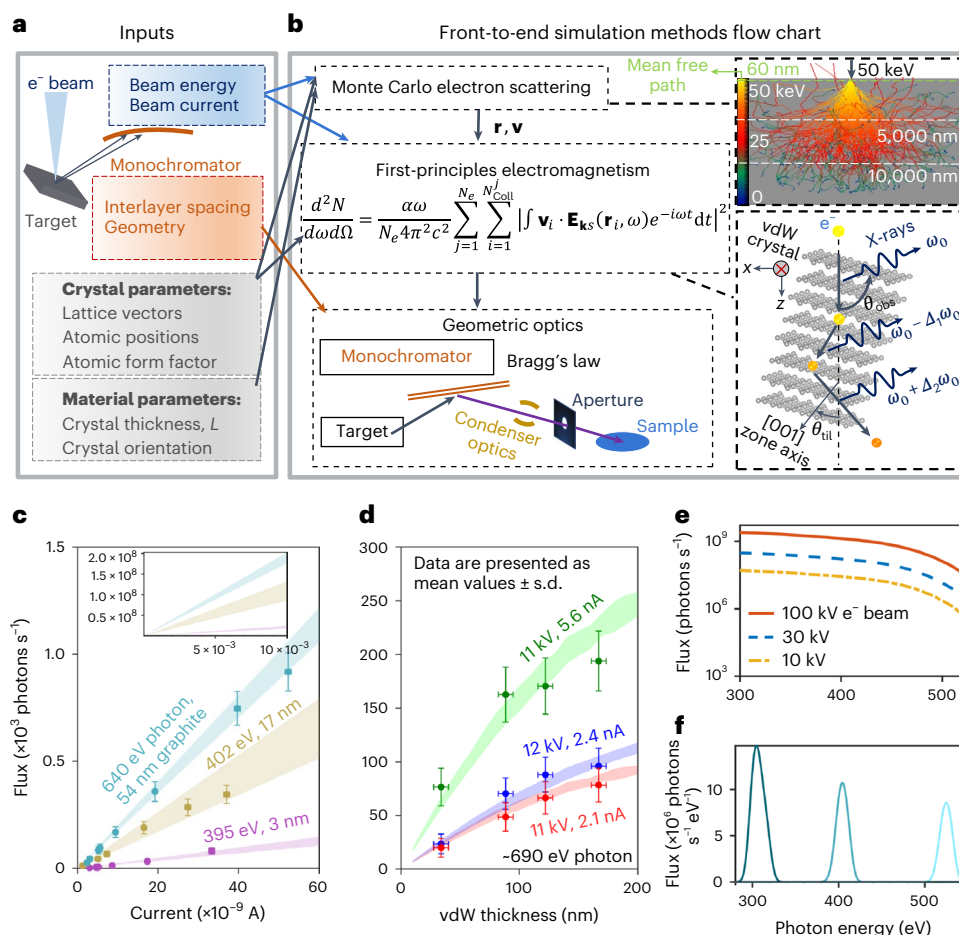


Fig. 2 | Theoretical framework to simulate photon flux in presence of electron scattering and experimental verification for flux scaling. **a**, Input parameters required for front-to-end (source-to-sample) simulation. **b**, Front-to-end theoretical framework for the simulation of flux falling on the sample. The top-right inset shows the Monte Carlo-simulated trajectories for a 50 keV electron beam. **c**, Flux (in photons s^{-1}) generated by the source (graphite) varies linearly with the incident electron-beam current, as detected by the EDS detector. The solid dots with the error bars denote the experimental data and the shaded region refers to the theoretical flux simulated using the numerical framework for varying azimuthal crystal tilt angles ϕ_{til} . The inset shows the scaled flux where the current is scaled up to 10 mA. The electron kinetic energies used to generate the 395 eV, 402 eV and 640 eV photon peaks are 3.8 keV, 4 keV and

10 keV, respectively. **d**, Variation in flux with increasing thickness of the crystal. **e**, Simulated flux for various photon energies falling on the sample for 10 mA current at three different incident electron energies. **f**, Spectrum of flux falling on the sample for three different photon energies showing narrow-bandwidth illumination, where the incident electron energy is 30 keV and the crystal thickness is 360 nm. For a given electron energy, the tilt angle θ_{til} of the crystal is adjusted (following equation (1)) to vary the emitted peak photon energy, and $\phi_{\text{til}} = 90^\circ$. The error bars represent the s.d. in the experimental results due to the Poisson statistical error in the photon-counting process of the EDS detector, given by $\sqrt{N_{\text{ph}}}$, where N_{ph} is the mean photon-counting rate, as well as the deviations in crystal thickness and incident beam current, estimated from three sample points.

physical parameters related to the incident electron beam, the target material and the monochromator. The arrows going from Fig. 2a to Fig. 2b show which inputs are needed to simulate the quantities listed in Fig. 2b.

Our framework begins by simulating electron scattering inside the crystal via Monte Carlo simulations (we use the CASINO software⁶¹). The top-right corner in Fig. 2b displays a Monte Carlo electron scattering simulation for a 50 keV electron beam interacting with a graphite target. The position of the electron at each collision event is saved along with its kinetic energy. The positional information $\mathbf{r}$ and velocity $\mathbf{v}$ of each trajectory are subsequently inputted into the photon number spectral density formula $d^2N/d\omega d\Omega$, which has been derived from Maxwell equations³⁰. Considering the crystal's geometry and detector orientation, X-ray attenuation due to material absorption is included at every electron collision point using the linear attenuation coefficient information obtained from ref. 62. More details about the X-ray absorption of the target material are given in Supplementary Section 5. Next, the resultant intensity (after factoring in the attenuation) for each electron

segment is summed over and finally averaged over the total number of electrons to obtain the final spectrum per electron. The number of electrons in the scattering simulation is increased until convergence is achieved in the output spectrum. The emitted spectrum from the target is then combined with an X-ray geometric optics setup consisting of a monochromator and condenser optics guided by an aperture to focus the X-rays onto the sample. Following Bragg's law, a narrowband X-ray beam is diffracted off the monochromator and focused onto the imaging sample using X-ray condenser optics.

Electron scattering suggests a decrease in intensity from the ideal case when there is no scattering. Therefore, using Monte Carlo electron scattering simulation, we obtain an effective interaction length (or PCB interaction length), which can be directly used to estimate the emitted flux. This method allows for a faster numerical computation and is used to determine the flux given in Fig. 2e,f and to generate the theory plots in Fig. 3. We define the PCB interaction length as the electrons' mean travel distance inside the crystal for which the deviation in the emitted photon energy from that of an unscattered electron is

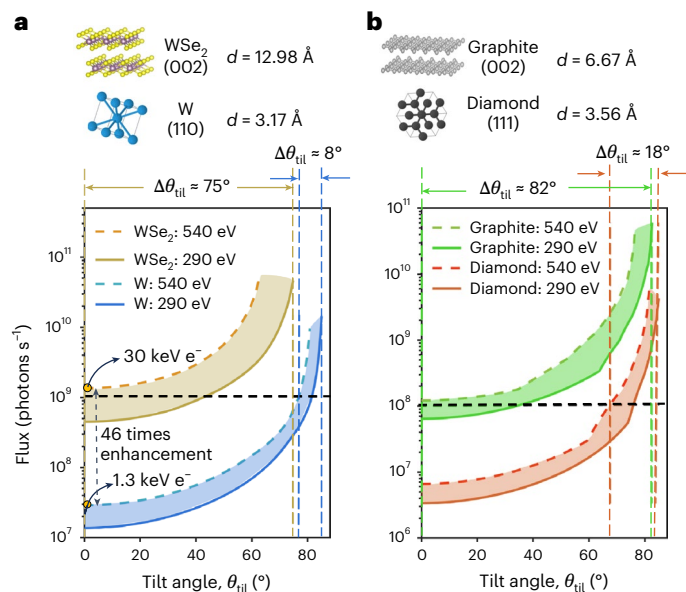


Fig. 3 | Advantages of vdW crystals over conventional crystals in generating tunable water-window X-rays. Crystals having a higher structure factor, deeper electron interaction length, longer mean free path and larger d spacing tend to produce larger flux within a narrow spectral band. Compared with W ($\Delta\theta_{\text{til}} \approx 8^\circ$), WSe₂ generates a much higher flux for a wide range of tilt angles ($\Delta\theta_{\text{til}} \approx 75^\circ$), as shown by the horizontal arrows in **a**. The case for diamond and graphite is similar, where diamond generates high flux within a very narrow tilt angle range $\Delta\theta_{\text{til}}$ of just $\sim 18^\circ$ and for graphite, $\Delta\theta_{\text{til}} \approx 82^\circ$, as shown in **b**. This limited range makes it challenging to precisely tune the photon energy across the entire water-window regime as the slightest mechanical errors can greatly affect the emitted photon energy. The incident electron energy along different curves is determined by equation (1) and is restricted to 1 MeV electron energy. Larger d spacing for vdW materials leads to reduced density, which leads to a larger mean free path and longer interaction length. In addition, a larger d spacing requires the incident electrons with comparatively higher kinetic energy to generate a fixed-energy photon, which, in turn, leads to even larger mean free paths and interaction lengths and eventually yields a higher flux. Moreover, a larger mean free path leads to a narrower bandwidth and thus contributes to higher narrow-bandwidth flux for vdW materials. The y axis in **a** and **b** corresponds to the flux integrated over 20 eV bandwidth, and the x axis denotes the crystal tilt angle θ_{til} . The crystal is oriented such that the reciprocal lattice vector corresponding to the dominant reflecting plane aligns with the z axis. In this figure, $\phi_{\text{til}} = 0^\circ$, $\theta_{\text{obs}} = 120^\circ$, $\phi_{\text{obs}} = 90^\circ$, the electron current is 10 mA and the solid angle considered is 0.03 sr.

less than 1%. The PCB interaction length, therefore, captures the electrons' average distance of travel during which the electrons continue to emit photons having energy within 1% of the desired photon energy. It should be noted that the PCB interaction length is typically far less than the electron penetration depth, where the latter term refers to the average distance travelled by electrons before they come to a complete stop. For example, the PCB interaction length of 50 keV electrons in a graphite crystal is 800 nm, which is two orders of magnitude smaller than the penetration depth in the graphite crystal (Fig. 2b, top right). The mean free path is defined as the average distance travelled by the electrons between two adjacent scattering events inside the crystal, for which the deviation in the emitted photon energy from that of an unscattered electron is less than 1%.

Using our theoretical framework, we predict the scaling of the output photon flux by increasing the crystal thickness and incident electron current. This prediction is confirmed through experimental results presented in Fig. 2c,d. As shown in Fig. 2c, the radiation flux demonstrates a linear increase with an increase in the input current. The measurement error in flux is considered for fluctuations in the electron current, surface thickness variations at different locations

($\sim 5\%$ uncertainty) and the error associated with the Poisson statistics of the photon-counting process. Due to equipment limitations, we were only able to go up to 50 nA in current, where the output X-ray flux is around 10^2 – 10^3 photons s^{-1} and the brightness is about 10^8 photons $\text{s}^{-1} \text{mm}^2 \text{mrad}^{-2}$ for 1% bandwidth. With rapid advances in few-photon and single-photon detection, such flux is already potentially useful in applications like X-ray absorption spectroscopy for elemental composition analysis^{63,64} and ghost imaging⁶⁵. However, extrapolating the current up to 10 mA would scale the output X-ray flux to the order of 10^8 photons s^{-1} (Fig. 2c, inset). Nevertheless, higher currents could result in a rise in the target temperature, leading to melting or damage. To assess this risk, we performed thermal simulations for a graphite crystal exposed to an electron beam. The simulation took into account heat conduction inside the graphite crystal and the black-body radiation mode of heat losses. The crystal was placed in a water-cooled copper holder, and the steady-state temperature at the graphite target centre was well below its melting temperature for a 100 keV electron beam at 10 mA current with a 1 mm spot size. The detailed thermal analysis procedure with more test cases is provided in Supplementary Section 6.

For high-current electron beams, space charge can also potentially affect our tunable water-window X-ray generation by modifying the energy and direction of the incident electron beam. Here the space charge refers to (1) space charge effects in electron-beam propagation and (2) space charge effects due to material charging. We have performed experiments and numerical simulations to show that even for a driving current of 10 mA, the effects of space charge are negligible. More details are given in Supplementary Section 7.

Increasing the thickness of the target crystal is another method to boost the X-ray intensity. Figure 2d depicts the increase in X-ray flux as the crystal's thickness rises from ~ 30 nm to ~ 170 nm. The thickness of the target crystal was measured with transmission electron microscopy (TEM) using the convergent beam electron diffraction technique^{66–68} (Supplementary Section 8). Combining our first-principles electromagnetism simulations with Monte Carlo simulations, we obtain the resulting flux emitted in direction $\hat{n}$ as

$$\frac{dN}{d\hat{n}} \propto (\omega |\mathbf{v} \cdot \mathbf{E}_{\text{gs}}|^2) i_c L^p, \quad (2)$$

where N denotes the number of emitted photons, ω denotes the peak photon frequency, $\mathbf{v}$ is the velocity, $\mathbf{E}_{\text{gs}}$ is the unit-cell scattering amplitude^{30,46}, i_c is the incident current, L is the sample thickness and $p \approx 3/4$ in our regime of study referencing the sublinear scaling of X-ray flux with increasing interaction length. Our numerical framework accurately predicts our experimental results, successfully capturing both the trend in flux variation and the precise flux values obtained from the experiment. In the context of electron scattering within vdW materials, this becomes particularly significant as it unveils an intriguing implication: the sublinear scaling of photon flux with electron interaction. It is important to highlight that as the crystal thickness increases beyond what was demonstrated in our experiments (Fig. 2d), the X-ray flux is expected to saturate. A more generalized version of our scaling law that accounts for this scenario is given in Supplementary Section 9.

Our proposed X-ray imaging setup utilizes a curved Si/W ellipsoidal mirror as the monochromator to enhance X-ray photon collection and diffraction towards the condenser optics. The ellipsoidal surface directs photons from the source to the second focus, where the condenser optics focuses the X-rays into a narrow spot on the sample. Our front-to-end simulations using our theoretical framework (Supplementary Section 2) showed a flux in the order of 10^8 photons s^{-1} on the sample surface generated from a graphite crystal with just 30 keV electron beam (Fig. 2e). Previous experiments have shown that such exposures are enough for imaging applications^{4,12,69–71}. The corresponding flux spectrum for three different photon energies is

shown in Fig. 2f, highlighting the narrow bandwidth of the radiation falling on the sample.

In particular, we can achieve fluxes in the order of 10^8 photons s^{-1} on the sample, comparable to the existing laboratory-based water-window X-ray sources^{4–7,9–12,70–72}. Unlike these sources, our output photon energy peak is continuously tunable across the water-window regime. On the basis of the manufacturers' specifications, we have considered an efficiency of 20% for the condenser optics (Wolter optics, zone plate or X-ray capillary tube) within the water-window regime and the reflectivity spectrum of the multilayer Si/W monochromator crystal is obtained from ref. 73. Imaging at different wavelengths offers greater-contrast imaging for different organic constituents within the biological cell because the X-ray transmission from different types of element changes with the energy of the X-rays. The flexibility to finely tune the X-ray energy, unlike the existing tabletop water-window X-ray imaging setups, allows our setup to participate in many applications requiring tunable X-rays. A narrow-bandwidth spectrum of the water-window X-rays, which can be achieved with our setup, is particularly valuable for studying the constituents of biological cell organelles⁷⁴. By restricting the illumination region of the ellipsoidal monochromator, we have control over the line shape of the spectrum that falls onto the sample. This control is easily achieved by incorporating a movable beam stopper (Supplementary Section 10).

X-ray flux as a function of the tilt angle of the crystal for vdW materials (tungsten diselenide (WSe₂) and graphite) and conventional crystals (tungsten (W) and diamond) is shown in Fig. 3. vdW materials offer an advantage over conventional crystalline materials because of their larger d spacing (interlayer spacing), since a larger d spacing reduces the density and leads to comparatively larger mean free path and interaction length. A longer mean free path leads to a narrower bandwidth of the emitted X-rays, increasing flux within this range. In addition, a larger d spacing allows a comparatively higher electron energy (within the tabletop electron guns) for water-window photon emission, which, in turn, leads to an even longer mean free path and interaction length of the electron, enabling the production of higher flux within a narrow spectral range. For example, W crystals only need 1.3 keV electrons to generate 540 eV photons at $\theta_{\text{til}} = 0^\circ$ and $\theta_{\text{obs}} = 120^\circ$, whereas WSe₂ crystals require 30 keV electrons to generate photons with the same energy at $\theta_{\text{til}} = 0^\circ$ and $\theta_{\text{obs}} = 120^\circ$. Since the PCB interaction length of 1.3 keV electrons in W crystals is around 1 nm and the PCB interaction length of 30 keV electrons in WSe₂ crystals is around 103 nm (103 times that for W), the flux of our tunable water-window X-ray sources generated from WSe₂ is significantly larger than that generated from W (Fig. 3a). A flow chart of this relation is shown in Supplementary Section 11. The advantage of a longer d spacing can be seen in a wider tilt-angle range, with WSe₂ producing higher flux compared with W, as indicated by $\Delta\theta_{\text{til}}$ between the blue and brown vertical dashed lines (Fig. 3a). W tends to produce a high flux of photons within a narrow tilt-angle range $\Delta\theta_{\text{til}}$ of only approximately 8° . Here we have considered the range for a higher-flux region greater than 10^9 photons s^{-1} . This limited range of tilt angles presents a challenge in terms of precisely tuning the energy of the emitted photons across the entire water-window regime. This is because even the slightest mechanical errors in the experiment can have a significant impact on the energy of the photons being produced, given the sensitivity of the system within such a narrow band of tilt-angle variability (Supplementary Section 17). The same is the case for graphite (vdW material) and diamond (conventional material) (Fig. 3b), where the tilt-angle range is considered for a flux regime greater than 10^8 photons s^{-1} . Different material parameters used in Fig. 3 are given in Supplementary Section 11. To get the flux values shown in Fig. 3, we have integrated across 20 eV spectral bandwidth of the X-ray peak emission. The X-ray peak spectrum was acquired from PCB radiation originating from the dominant reflecting plane with the highest structure factor, as indicated at the top of Fig. 3 next to each material. The electron energy used to generate the theoretical curves in Fig. 3 is

determined using equation (1), with a limitation set at 1 MeV to suit the capabilities of tabletop electron sources. As X-ray absorption within the material is dependent on geometric conditions, it has been ignored for the curves presented in Fig. 3. For instance, a tailored flat electron beam^{75–78} focused near the crystal's edge and penetrating through the material close to the edge minimizes X-ray absorption and the generated flux would depend on the electron's PCB interaction length.

Discussion

vdW materials are an attractive platform for water-window X-ray sources based on free-electron-driven quantum materials. The vdW X-ray source's frequency is dynamically tunable, scanning the wavelengths within the entire water-window regime, unlike traditional X-ray tubes whose output X-ray peaks are fixed at the characteristic frequencies of the anode material. The proposed vdW-material-based free-electron-driven source is simpler to use and can be operated on a tabletop setup. Furthermore, it requires neither high-intensity lasers, as in high-harmonic generation⁷⁹ and laser plasma sources⁸⁰, nor highly relativistic electrons, as in undulator-based X-ray sources^{81,82}. Thus, vdW-based water-window continuously tunable X-ray sources can be a useful tool in X-ray microscopy and spectroscopy applications. Continuous photon energy tunability is also useful in NEXAFS spectroscopy^{10,11,83–87} including the detection of elements with their absorption edges in the water window^{74,88}, which is the case for some essential elements forming organic matter (Supplementary Section 13). Recent breakthroughs in shaping free-electron wavepackets^{75,89–99}, laser-driven electron acceleration¹⁰⁰ and X-ray generation into waveguide modes¹⁰¹ could lead to greater control over and enhancement of the output radiation from our source.

With rapid advances in few-photon and single-photon detection, our experimentally demonstrated flux of 10^2 – 10^3 photons s^{-1} (Fig. 2c,d) is already potentially useful in applications like X-ray absorption spectroscopy for elemental composition analysis^{63,64} and ghost imaging⁶⁵. With a further scaling of flux, this compact continuously tunable water-window X-ray source can be potentially useful for the accurate detection of Ca-rich DG in blood platelet cells and helps in the early detection of osteoporosis. The standard method currently used to assess DG deficiency in blood platelets is using TEM. The correct identification of DG would be important to assess the status of platelets and DG-related diseases. However, due to the electron-density-based contrast mechanism in TEM, other granules such as α -granules might cause false DG detection. Using X-ray transmission microscopy instead minimizes false DG detection of human platelets. DG can be identified by exploiting its high Ca content. Two X-ray transmission images are taken, one below the Ca L-absorption edge and one above (post-edge region). A Ca map can be obtained from these two images, accurately revealing the DG regions. This method under scanning transmission X-ray microscopy is successfully shown in synchrotron facilities using tunable water-window X-rays³. Although our source may not attain the ultrahigh photon fluxes of synchrotron sources, our studies show that the photon flux and other properties of our X-ray source are sufficient to generate high-contrast images for DG detection. This makes our source highly complementary to synchrotron sources: our source is even highly advantageous in which compactness is important but the ultrahigh photon fluxes of synchrotrons are not necessary. To directly validate the usefulness of our source in applications like biological imaging, we have studied phantom models of blood platelet cells in Supplementary Section 3, also presented in the inset of Fig. 1b (showing 2,000-fold contrast enhancement). vdW-based free-electron-driven sources provide a complementary tool to the present the TEM-based technique for conducting such studies on thin samples in a laboratory environment. Similar Ca mapping in mineralized tissues by absorption-difference imaging using two X-ray energies near the Ca L-absorption edge and combining it with NEXAFS spectroscopy analysis provides a useful tool to quantify Ca concentration in tissues².

Other potential applications for our continuously tunable, narrowband water-window X-ray source includes NEXAFS spectroscopy for organic materials^{84,85}, chemical mapping of biological samples⁷⁴ and various non-biological applications such as in the microchemical analysis of inorganic materials (paleobotany¹⁰²). NEXAFS, in particular, is used to investigate the structure of intermolecular bonds of polymers⁸⁴ by probing the electronic transition from the core level to the unoccupied states. Due to the distinct core binding energies that each element possesses, the NEXAFS spectra can reveal elemental information. NEXAFS spectroscopy studies near the absorption edges of various elements present in organic matter, such as organic thin films, help to determine the dynamic electronic properties⁸⁵. Such applications can be performed by our proposed X-ray imaging setup by replacing the CCD camera with a spectrometer.

Our results reveal that vdW materials have an undisputed edge over conventional materials due to the larger d spacing of vdW materials, which provides greater interaction length and longer mean free path, ultimately resulting in a higher flux within a narrower spectral bandwidth. The interaction length increases because of the reduction in crystal density since the volume increases as the d spacing of the crystal increases. The scaling of PCB intensity is a critical factor in many X-ray applications and understanding the fundamental laws governing this process is of utmost importance. Through this research, we have uncovered the fundamental scaling laws of PCB intensity, which show that it scales linearly with current and sub-linearly with thickness. We demonstrated these scaling laws through theoretical predictions and experimental validation, which has not been previously demonstrated. This discovery has significant implications for the field, as it opens up new possibilities for controlling and optimizing the intensity of PCB emissions from X-ray sources. Our findings have the potential to impact a wide range of industries that rely on X-ray technology, from medical imaging to materials analysis and beyond.

In our regime of study, the shifts in the actual outgoing photon energies from their predictions by classical theory (equation (1))—known as quantum recoil^{42,43}—are less than 3.5%. It should be noted that when the photon energy shift due to quantum recoil is small, the correction in the actual generated X-ray intensity from its classically predicted value is also small¹⁰³. This explains why our theoretical and experimental results are in good agreement, even though we have used a theoretical framework that does not capture quantum recoil.

It should also be noted that our tunable water-window X-rays can be generated by electrons with energies ranging from a few kiloelectronvolts to a few megaelectronvolts. However, the electron energies that we used to experimentally demonstrate our tunable water-window X-ray generation are below 10 keV. Lower-energy electrons (below 10 keV) are appealing in allowing the desired X-rays to be generated with lower-voltage power supplies, which could lead to lower costs and a smaller footprint compared with a setup involving higher-voltage power supplies. We note, however, that it can also be advantageous to operate our tunable water-window X-ray source using higher-energy electrons, which could lead to higher fluxes and brightness. The effects of electron-beam divergence and energy spread on our tunable water-window X-ray generation are studied in Supplementary Section 12. The output characteristics of our tunable X-ray sources, across various parameters such as the energy of the electron beam, the thickness and type of the material, and the size of the beam spot are explored in Supplementary Section 14. A comparison between our tunable water-window X-ray and existing laboratory water-window X-ray sources is given in Supplementary Section 15.

In conclusion, we present a versatile, tabletop X-ray source in which continuously tunable water-window X-rays are generated from free-electron-driven vdW structures. We present a truly predictive theoretical framework that combines first-principles electromagnetism and Monte Carlo simulations to accurately predict the output photon flux and brightness of free-electron-driven quantum materials

in absolute numbers. Using this framework, we theoretically obtain and experimentally reveal the fundamental scaling laws of the emitted photon flux. We show that the tunable photon flux increases linearly with current and sub-linearly with crystal thickness. Importantly, our theoretical framework also enabled the front-to-end simulation of water-window imaging setups, enabling us to show the feasibility of fluxes exceeding 10^8 photons s^{-1} on the sample, as verified by the rigorous simulation based on experimentally achieved fluxes of approximately 10^3 photons s^{-1} using ~50 nA current. Such fluxes have been shown to be sufficient for a broad range of laboratory-based imaging and spectroscopy applications, but our setup also benefits from further enhancements in contrast enabled by the photon energy peak being continuously tuned across the water-window regime. We demonstrate the ability to continuously vary the photon energy peak across the entire water-window regime by varying the tilt angle of the crystal and/or the driving electron energy. Our framework also reveals the distinct advantages of vdW materials over conventional crystalline materials in our scheme, resulting in higher photon fluxes with narrower bandwidths. Our results bring a highly unique advantage of synchrotron-based water-window X-ray sources—namely, continuous photon energy tunability—to the tabletop scale, and should unlock exciting prospects in high-resolution imaging and spectroscopy for medical and biological applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-024-01547-3>.

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Methods

The X-ray measurements were performed using a JEOL 7800 field-emission scanning electron microscope. The emitted X-ray spectra were measured using a silicon-drift EDS detector in the field-emission scanning electron microscope. The EDS detector was calibrated to measure the X-ray photon energies with an accuracy within ± 1 eV by measuring the K_{α} peaks of C, N, B, O, F and Si (Supplementary Section 16). The solid angle of the detector is 0.04 sr and the collection angle of the detector is around 13° . The collection angle would broaden the bandwidth of our tunable water-window X-ray sources by about 7 eV. The background radiation of the measured spectra was subtracted using NIST DTSA-II software^{104,105}. The details of the methods used to analyse the experimental data are described in Supplementary Section 4.

In our sample preparation method, graphite nanoflakes are mechanically exfoliated onto silicon substrates (with a 285 nm SiO_2 film) and transferred onto gold grids using the wet-transfer method. The gold grid is held by a copper TEM grid holder.

Theory and simulations for PCB intensity are discussed in Supplementary Sections 1 and 2. Water-window X-ray imaging setup and front-to-end simulations using our theoretical framework are provided in Supplementary Section 2. Convergent beam electron diffraction thickness measurements are discussed in Supplementary Section 8.

Data availability

Data for Figs. 1–3 are available in the Supplementary Information. Source data are provided with this paper. All other data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

Code availability

All code that supports the plots within this paper is available from the corresponding author upon reasonable request.

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Author contributions

N.P. led the project. N.P. and S.H. designed and performed the electron microscopy experiments. R.D. and Z.L. prepared the samples. N.P., S.H., M.G., Q.Z. and C.B. analysed the experimental results. N.P. developed the theory and performed the simulations. N.P., S.H. and L.J.W. wrote the paper, with inputs from all other authors. L.J.W. conceived the idea and supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

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